PARAMETRIZING A POPULATION BALANCE EQUATION

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Abstract. Surely this cannot be that difficult.

1. Introduction

A plethora of quantities arise in this work, so we endeavour to consistently indicate their nature at a glance. All constants $\|\cdot\|, \langle\cdot\rangle, |\cdot| \in \overline{\mathbb{R}^+}$ are non-negative, and enclosed in $\|\cdot\|$ or $\langle\cdot\rangle$ without meaning (there are no norms or inner products here), or $|\cdot|$ only when taking the absolute value of $\cdot$. All *italic* quantities are dimensionless, to ease the unavoidable chore of tracking units. Undefined notation adheres to the Digital Library of Mathematical Functions [1].

A number $\mathrm{n}(\mathrm{t},\mathrm{L})$ (m^{-4}) of particles of length $\mathrm{L}\,(\mathrm{m})$ grow within each unit of reactor volume $\mathrm{V}\,(\mathrm{m}^3)$ as seconds $\mathrm{t}\,(\mathrm{s})$ pass. Conserving mass, the Population Balance Equation is

$$\begin{aligned} \partial_{\mathrm{t}}(\mathrm{Vn}(\mathrm{t},\mathrm{L})) &= -\mathrm{G}^+\partial_{\mathrm{L}}(\mathrm{Vn}(\mathrm{t},\mathrm{L})) \\ \mathrm{n}(0,\mathrm{L}) &= 0 \\ \mathrm{n}(\mathrm{t},0) &= \mathrm{B}^+/\mathrm{G}^+ \end{aligned} \tag{1.1}$$

where $\mathrm{B}^+\,(\mathrm{m}^{-3}\mathrm{s}^{-1})$ and $\mathrm{G}^+\,(\mathrm{m\,s}^{-1})$ are particle birth (nucleation) and growth rates. These relate empirically to supersaturation $S > 1$:

$$\begin{aligned} \text{birth} \quad \mathrm{B}^+ &:= \|\mathrm{B}\|\,(S-1)^{+|b|} \\ \text{growth} \quad \mathrm{G}^+ &:= \|\mathrm{G}\|\,(S-1)^{+|g|} \end{aligned} \tag{1.2}$$

where $(\#)^+ := \max(0,\#)$ denotes the positive part of $(\#)$. In experiments, critical supersaturation $S = 1$ might occur many times, but we cannot allow this. We must insist that

$$\text{supersaturation} \quad S \in (1, \sup S] \quad \text{at all times} \quad \mathrm{t} > \mathrm{t}^* \tag{1.3}$$

for finite supersaturation $\sup S \in \mathbb{R}$ and critical time $\mathrm{t}^* \in \mathbb{R}$ at which S first exceeds 1. This is not limiting per se, but it does mean that each critical instant in an experiment theoretically begins a new experiment. Throughout this work, an asterisk always denotes the critical instant $t = S - 1 = 0$

Given lengthscale $\|\mathrm{L}\|\,(\mathrm{m})$ and timescale $\|\mathrm{t}\|\,(\mathrm{s})$, dimensionless prefactors are defined as

$$\begin{aligned} \text{birth} \quad \|B\| &:= \|\mathrm{B}\|\,\|\mathrm{L}\|^3\,\|\mathrm{t}\| \\ \text{growth} \quad \|G\| &:= \|\mathrm{G}\|\,\|\mathrm{L}\|^{-1}\,\|\mathrm{t}\| \end{aligned} \tag{1.4}$$

together with dimensionless

$$\begin{aligned} \text{time} \quad t &:= \|\mathrm{t}\|^{-1}\,(\mathrm{t}-\mathrm{t}^*) \\ \text{volume} \quad V(t) &:= \|\mathrm{L}\|^{-3}\,\mathrm{V} \end{aligned} \tag{1.5}$$

This work aims to elucidate the dimensionless empirical birth and growth parameters $\|B\|, |b|, \|G\|, |g|$.

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2. The Crystal Size Distribution (CSD)

The Population Balance Equation conveys a symmetry between synthetic 'time' T and 'length' L defined as

$$S(t) := \max(S, 1) \tag{2.1a}$$

$$T := \|\mathrm{L}\|^{-1} \int_0^{\mathrm{t}} \mathrm{G}^+ \,\mathrm{dt} = \int_0^t \|G\| \left(S(\dot{t}) - 1\right)^{|g|} \mathrm{d}\dot{t} \tag{2.1b}$$

$$L := T - \|\mathrm{L}\|^{-1} \mathrm{L} \tag{2.1c}$$

These forms are chosen to render the PBE linear and invariant to exchanging T and L. This symmetry manifests in the particle count $N(T)f(T,L)$ defined as

$$N(\mathbb{R}^-) := \{0\} \tag{2.2a}$$

$$N(T) := \int_0^{\mathrm{t}} \mathrm{B}^+ \mathrm{V} \,\mathrm{d\dot{t}} = \int_0^t \|B\| \left(S(\dot{t}) - 1\right)^{|b|} V(\dot{t}) \,\mathrm{d}\dot{t} \tag{2.2b}$$

$$f(T,L) := N(T)^{-1} \|\mathrm{L}\|^{-1} \mathrm{Vn(t, L)} \tag{2.2c}$$

Substituting these variables in Population Balance Equation (1.1) immediately yields

$$\begin{aligned} \partial_T N(T) f(T,L) &= 0 \\ N(T) f(T,L) &= \partial_t N(T) / \partial_t T = \partial_T N(T) \end{aligned} \tag{2.3}$$

These two lines combine to establish

$$\begin{aligned} f(T,L) &= N(T)^{-1} \partial_L N(L) \\ N(T)^{-1} N(L) &= \int_0^L f(T, \dot{L}) \,\mathrm{d}\dot{L} \end{aligned} \tag{2.4}$$

The Crystal Size Distribution (CSD) at 'time' T is $N(T)^{-1}N(L)$, its probability density $f(T,L)$, and its moments

$$M_m(T) := \int_0^T f(T,L)(T-L)^m \,\mathrm{d}L = N(T)^{-1} m \int_0^T N(\dot{T}) M_{m-1}(\dot{T}) \,\mathrm{d}\dot{T} \tag{2.5}$$

The number of particles born by 'time' T is $N(T)$, and the number of particles whose 'length' does not exceed L is $N(L)$. This is the symmetry conveyed by the PBE: 'time' T and 'length' L are effectively the same thing. Like a growing child, a crystal equates how old it is with how big it is.

The CSD $N(T)^{-1}N(L)$ is the fraction of particles at 'time' T whose 'length' does not exceed L, which is the same as the fraction of particles at 'time' T born by 'time' L.

3. Taylor Approximation

This Section determines the CSD from supersaturation $S(t)$ and volume $V(t)$, expanded about $t = \bar{t}$ as

$$\begin{aligned} \ln(S(t) - 1) \;&= \|\bar{\sigma}\| + \|\bar{\sigma}'\| (t - \bar{t}) + \|\bar{\sigma}''\| \frac{(t-\bar{t})^2}{2} + O\left(\|\bar{\sigma}'''\| \frac{(t-\bar{t})^3}{6} \right) \\ \|\bar{\sigma}^{(n)}\| &:= \partial_{\bar{t}}^n \ln(S(\bar{t}) - 1) \\ V(t) \;&= \left\|\bar{V}\right\| + \left\|\bar{V}'\right\| (t - \bar{t}) + \left\|\bar{V}''\right\| \frac{(t-\bar{t})^2}{2} + O\left(\left\|\bar{V}'''\right\| \frac{(t-\bar{t})^3}{6} \right) \\ \|\bar{V}^{(n)}\| &:= \partial_{\bar{t}}^n V(\bar{t}) \end{aligned} \tag{3.1}$$

The partial derivatives in the second line are normally provided by an exogenous model of supersaturation, our sign convention dictating $\|\bar{\sigma}^{(n)}\| \geq 0$.

Completing the square

$$\begin{aligned} \ln(S(t) - 1) \;&= \langle \bar{s} \rangle + \|\bar{\sigma}''\| \frac{\left(t - \|\bar{t}\|\right)^2}{2} + O\left(\|\bar{\sigma}'''\| \frac{(t-\bar{t})^3}{6} \right) \\ \|\bar{t}\| &:= \bar{t} - \|\bar{\sigma}'\| \, \|\bar{\sigma}''\|^{-1} \\ \|\bar{s}\| &:= \|\bar{\sigma}\| - 2^{-1} \|\bar{\sigma}'\|^2 \, \|\bar{\sigma}''\|^{-1} \end{aligned} \tag{3.2}$$

so

$$
\begin{aligned}
V(t) \; &= \|\bar{v}\| + \|\bar{v}'\| \, \|\bar{\sigma}''\| \, (t - \|\bar{t}\|) + \|\bar{v}''\| \, \|\bar{\sigma}''\|^2 \, (t - \|\bar{t}\|)^2 + O\left(\left\|\bar{V}'''\right\| \frac{(t-\bar{t})^3}{6} \right) \\
\|\bar{v}''\| &:= \left\|\bar{V}''\right\| \|\bar{\sigma}''\|^{-2} \, 2^{-1} \\
\|\bar{v}'\| &:= \left\|\bar{V}'\right\| \|\bar{\sigma}''\|^{-1} - 2 \, \|\bar{v}''\| \, \|\bar{\sigma}'\| \\
\|\bar{v}\| &:= \left\|\bar{V}''\right\| \|\bar{\sigma}''\|^{-2} \, 2^{-1}
\end{aligned}
\tag{3.3}
$$

Asymptotic 'time' T is determined as

$$
\begin{aligned}
T &= \|G\| \int_0^t (S(\dot{t}) - 1)^{|g|} \mathrm{d}\dot{t} = \|G\| \int_0^t \exp\left(|g| \ln(S(\dot{t}) - 1) \right) \mathrm{d}\dot{t} \\
&= \|G\| \int_0^t \left(1 + |g| \ln(S(\dot{t}) - 1) + O(|g|^2) \right) \mathrm{d}\dot{t}
\end{aligned}
\tag{3.4}
$$

References

1. *NIST Digital Library of Mathematical Functions*, `https://dlmf.nist.gov/`, Release 1.2.6 of 2026-03-15, F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.